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Team 3

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Group Case 2: Metabical Pricing, Packaging, and Demand Forecasting for a New Weight-Loss Drug

Rutendo Q1. How does Metabical compare to current weight-loss options?

Metabical stands out from existing weight-loss options because it addresses a group that has largely been overlooked by the pharmaceutical industry: adults who are moderately overweight, with a BMI between 25 and 30. Most prescription weight-loss drugs are designed for obese or severely obese individuals. Products such as Meridia and Xenical can also cause serious side effects, including heart, liver, kidney, and gallbladder complications. Metabical offers a safer alternative because its ingredients are administered in lower doses and generally produce less severe side effects.

Compared with over-the-counter products, Metabical also appears more reliable. Alli, for example, is FDA-approved but must be taken three times a day and may cause gastrointestinal discomfort and possible liver-related concerns. Metabical is taken only once daily, making it more convenient. Its effectiveness is also supported by clinical trials. Participants with BMIs between 25 and 28 lost an average of 15 pounds, while those with BMIs between 28 and 30 lost approximately 26 pounds. These results were much better than those achieved by participants who received a placebo.

Metabical also competes with programs such as Weight Watchers, Jenny Craig, diet plans, and gym memberships. These alternatives can support healthier lifestyles, but their success often depends heavily on the individual's motivation and consistency. Jenny Craig can also cost between \$1,000 and \$1,600 for three months, making Metabical a potentially more affordable option. An additional benefit is that Metabical's side effects discourage users from eating high-fat and high-calorie foods, which may help them develop healthier habits.

However, Metabical is not without challenges. It requires a prescription, may not be covered by insurance, and could cost between \$225 and \$450 for the complete 12-week program. Overall, I believe Metabical has a strong competitive position because it combines medical supervision, convenience, proven effectiveness, and relatively manageable side effects. Its success will depend on whether consumers believe these benefits justify paying for the treatment out of pocket.

NIA- Q2. What are the pros and cons of the forecasting methods presented by Printup? If you had to estimate the demand for this product, how would you go about it? What would your demand (unit) forecast look like for the first 5 years? (I want numbers – not just words).

In order to estimate product demand, Approach 2 is the most effective, because it is focused on consumers who have already expressed interest (12%, p.6) and willingness to seek a prescription. The case states that 60% of first time users purchase a second supply and 20% of those consumers will purchase the final supply, which means the forecast should include repeat purchases. The American adult population also increases slightly annually. We should calculate the overweight population each year as opposed to the fixed value 78.2 million.

Approach 2 is the most promising approach because 9.38 million people have expressed interest in seeking a prescription which indicates product demand. The forecast incorporates the case's 10% to 30% penetration assumptions and 60%/20% repeat purchases, producing a five-year forecast of approximately 16.6 million units.

	Pros	Cons
Approach 1: Overweight individuals who are actively trying to lose weight.	The market is very large and essentially easy to segment and target. The use of a conservative market can reduce the risk of overestimating demand.	The market is full of potential consumers although some may prefer dieting, exercising, weight loss surgery, drugstore products, or other remedies. The demand forecasting will be riskier due to Metabical being a new prescription drug with fewer side effects. Approach 1 also assumes that only 15% of people trying to lose weight would be comfortable with weight loss medication.
*** Approach 2: Individuals ready to request a prescription.	Purchase intent can be measured more accurately with the second approach. The consumer research identifies 12% are ready to go to a healthcare provider, which allows various types of consumers to be targeted. The repeat purchase assumption 60%/20% from the clinical trials proves to be effective. Approach 2 is not super conservative in comparison to Approach 1 and Approach 3 is very targeted.	The survey responses do not guarantee purchase, while assuming that potential consumers do seek a prescription from their healthcare professional. Demand forecasting can potentially be overestimated due to insurance coverage, pricing, and other drug recommendations. The 12% survey response can overestimate demand once consumers are aware of pricing.
Approach 3: The Ideal Consumer	The market is very niche for the target consumer. Metabical advertising campaigns and promos will be easy to design and execute with the target consumer in mind.	The demand forecast projects 30% penetration in year 1 and increasing by 20% in year 5. The target consumer does not include male consumers and other demographics which are a potential market.

Potential Users Before Penetration

*230 million * 34% = 78.2 million overweight individuals*

Approach 1 (People trying to lose weight 35% and potential users 15%)

78.2 million * 35% * 15%
= 4.11 million consumers

Approach 2

78.2 million * 12% (potential users)
= 9.38 million consumers

Approach 3

4.3 million consumers
college-educated women ages 35–65 with a BMI between 25 and 30

Demand Unit Forecast

	US Population	34% Overweight	12% Potential	Penetration	New Users	60% - Repeat Purchase	20% - Repeat Purchase	Total Units
Year 1	230 M	78.2 M	9.38 M	10%	938,400	563,040	112,608	1,614,048
Year 2	233 M	79.22 M	9.51 M	15%	1,425,960	855,576	171,115	2,452,651
Year 3	235 M	79.9 M	9.59 M	20%	1,917,600	1,150,560	230,112	3,298,272
Year 4	238 M	80.92 M	9.71 M	25%	2,427,600	1,456,560	291,312	4,175,472
Year 5	240 M	81.6 M	9.29 M	30%	2,937,600	1,762,560	352,512	5,052,672

HEVEN-Q3. What considerations should be taken into account when making decisions about the package count? What package size would you recommend and why?

A four-week supply containing 28 tablets represents the most strategic option. Key factors influencing packaging decisions include out-of-pocket affordability, patient compliance, consumer motivation, repeat purchase rates, and overall margins. While a full twelve-week supply guarantees complete regimen adherence and removes renewal requirements, it imposes a significant initial financial burden. This distinction is critical given that Metabical will largely rely on cash-pay customers due to limited third-party insurance coverage. Offering a four-week count balances financial accessibility and therapy continuation by minimizing the starting cost while providing sufficient dosage to demonstrate efficacy. Although requiring patients to obtain two subsequent refills introduces potential therapy drop-offs, lowering the initial price barrier is vital for driving trial among first-time users. Consequently, packaging 28 tablets per bottle, requiring three monthly fulfillments for the full regimen, serves as the optimal configuration.

JULIA Q4. What pricing strategy would you suggest Printup explore? What are the advantages and disadvantages of each strategy? What price would you recommend?

Printup should explore a **value-based pricing strategy** for Metabical. Metabical is a new prescription drug with proven results, fewer side effects, and the convenience of being taken once a day, the price should reflect the value that customers receive from the product.

A **cost-based pricing strategy** would be simple because Printup could calculate the cost of producing the drug and then add a profit margin. However, this approach does not consider what customers are willing to pay or the benefits they receive from Metabical. A **competitor-based strategy** would allow Printup to compare Metabical with other weight-loss options, but it could be difficult

because many of those alternatives are not directly comparable to a prescription drug. Pricing based only on competitors could also cause Printup to undervalue Metabical.

With **value-based pricing**, Printup can charge based on the benefits and results customers believe Metabical provides. The main advantage is that it gives Printup more flexibility to earn a strong return while still positioning Metabical as a premium product. The disadvantage is that customers will have to pay much of the cost themselves because insurance coverage may be limited. If the price is too high, some potential customers may decide not to use the product.

I would recommend pricing the **12-week treatment at \$225**, which is the lower end of the \$225-\$450 range given in the case. This price makes Metabical more affordable for customers paying out of pocket while still reflecting the value of a prescription treatment with proven results. I think \$225 would give Printup a good balance between making the product accessible and maintaining its value in the market.

Bottom Line: **Value-based pricing + \$225 for the 12-week treatment.**

Haotian **Q5. What impact does your pricing strategy have on profitability? What is the ROI over the first 5 years for each of the pricing strategies identified?**

We recommend pricing Metabical at \$225 for the full 12-week treatment, equivalent to \$75 per four-week package. This accessible price could encourage product trial, improve treatment completion, build market share, and reduce the financial burden on customers without insurance coverage. It could also help Metabolic Establishment establish itself as the leading prescription drug for moderately overweight consumers.

Profitability Calculation

The baseline forecast predicts sales of 7,061,460 four-week packages over the first five years.

- Manufacturer revenue per package: \$50.00
- Variable cost per package: \$25.20
- Contribution margin per package: \$24.80
- Total five-year contribution margin: \$175.12 million
- Five-year fixed costs: \$6 million
- Initial marketing expense: \$23 million
- Five-year operating profit: \$146.12 million
- Initial R&D investment: \$400 million
- Return after R&D investment: -\$253.88 million
- Five-year ROI: -63.5%

Pricing Strategy Comparison

12-Week Treatment Price	Five-Year Operating Profit	Return After R&D	Five-Year ROI		
\$225	\$146.12 million	-\$253.88 million	-63.5%		
\$375	uncertainties Furbrig!	brightfarsisht!	crylich! breathe \$381.48 million	-\$18.52 million	-4.6%
\$450	\$499.20 million	\$99.20 million	24.8%		

Conclusion

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The \$225 price is a strategically appropriate launch price because it balances affordability with Metabical's differentiated value. At this price, each package generates a positive contribution margin of \$24.80, resulting in a five-year operating profit of \$146.12 million.

Although the calculated ROI is negative after including the full \$400 million R&D investment, this result is largely driven by the substantial sunk development cost and the conservative demand forecast. The comparison also assumes that demand remains the same at every price, which may overstate the profitability of the higher-priced options. In reality, a lower price could increase prescription fulfillment, treatment completion, physician adoption, and customer word-of-mouth.

Therefore, the \$225 price provides a reasonable balance between customer accessibility and profitability. It allows Metabical to maintain a premium over Alli while encouraging broader adoption and helping CSP establish the product as the leading prescription treatment for moderately overweight consumers.